Linear Programming Approach to the Graph Square Problem

Edward Xu

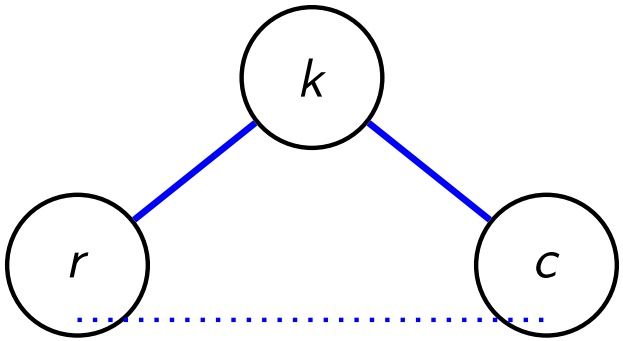
University of Melbourne

Supervisor: Dr Christopher Duffy

October 21, 2026

Square of a Graph

Two-step connections

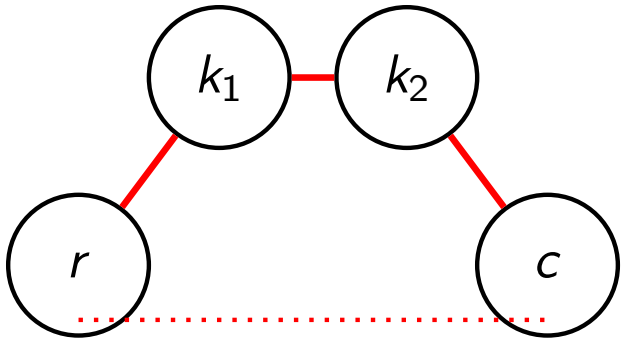


two steps

$$(A^2)_{rc} = \sum_{k=1}^{n} A_{rk} A_{kc}.$$

Counts walks of length two from r to c.

Three-step connections

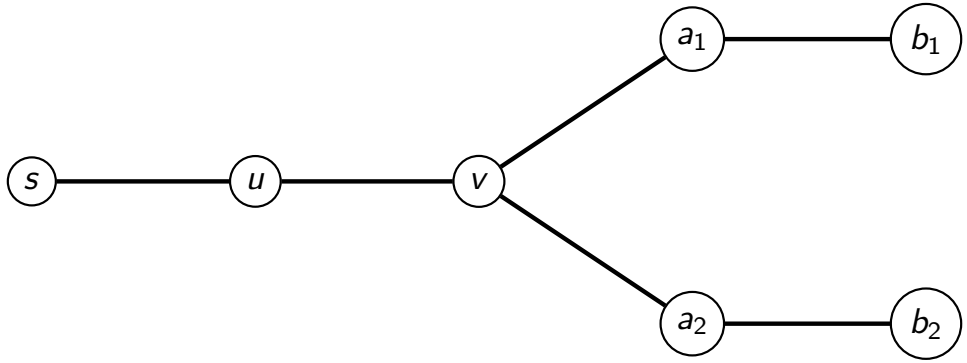


three steps

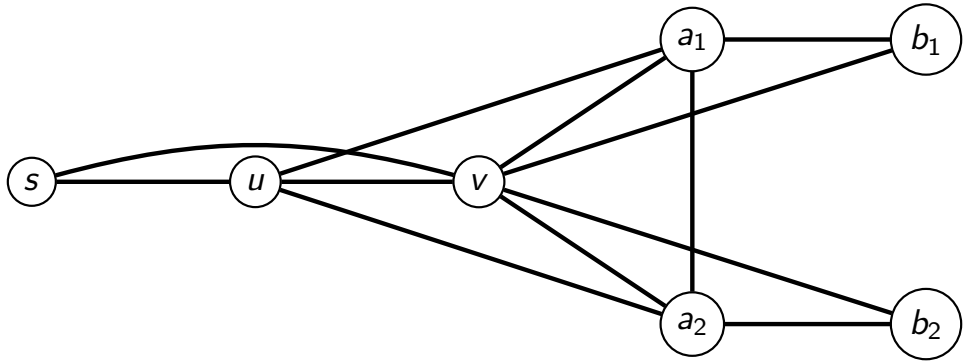
$$(A^3)_{rc} = \sum_{k=1}^{n} A_{rk} (A^2)_{kc}.$$

Counts walks of length three from r to c.

Graph Root



Graph Square or Square of a Graph



The Graph Square Problem

Problem (SQUARE)

Instance: *A finite, simple, undirected graph* $G = (V, E)$.

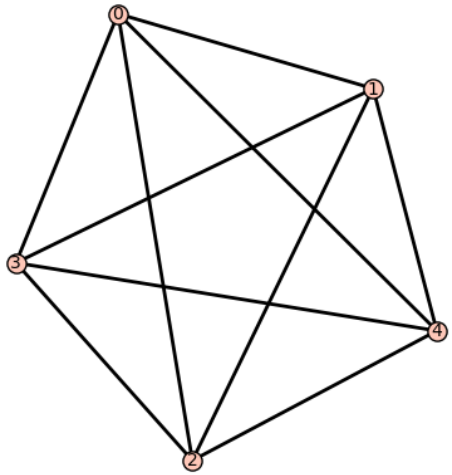
Question: *Does there exist a finite, simple, undirected graph* H *such that*

$$G = H^2?$$

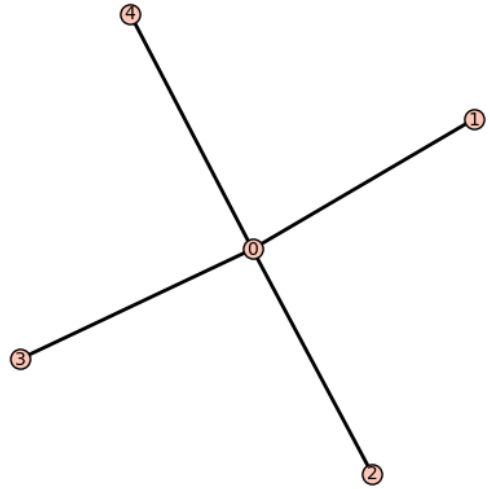
Let the **graph root** $H = (V, E_H)$ be a finite, simple, undirected graph, where V is the vertex set with $|V| = n$, and $E_H \subseteq \{\{u, v\} \mid u, v \in V, u \neq v\}$ is the edge set. The **square of a graph** H, denoted by $H^2 = (V, E_{H^2})$ is the graph on the same vertex set with edge set

$$E_{H^2} = \{uv \mid u \neq v, 1 \leq d_H(u, v) \leq 2\}. \tag{1}$$

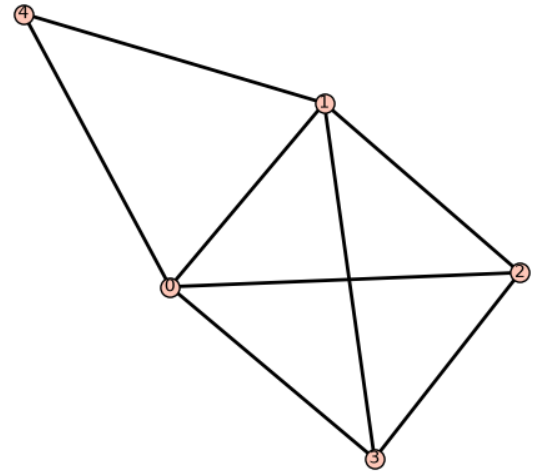
When $G = H^2$ we say that H is a **root** of G.



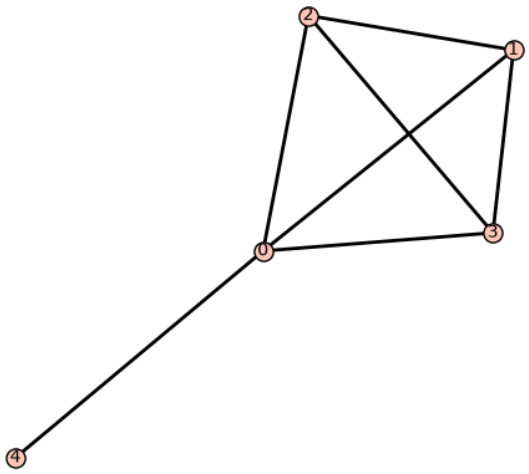
(a) K_5, which is a root of itself.



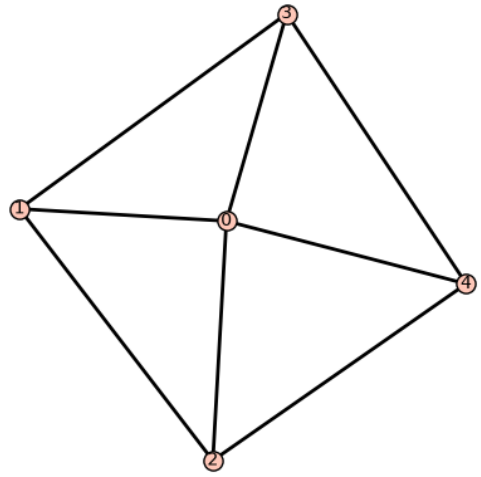
(b) $K_{1,4}$, a root of K_5.



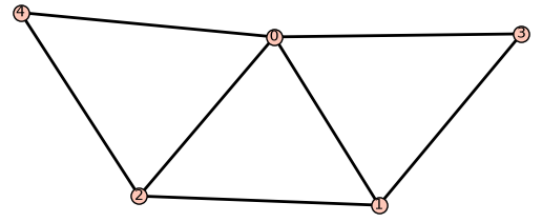
(c) A root of K_5.



(d) A root of K_5.

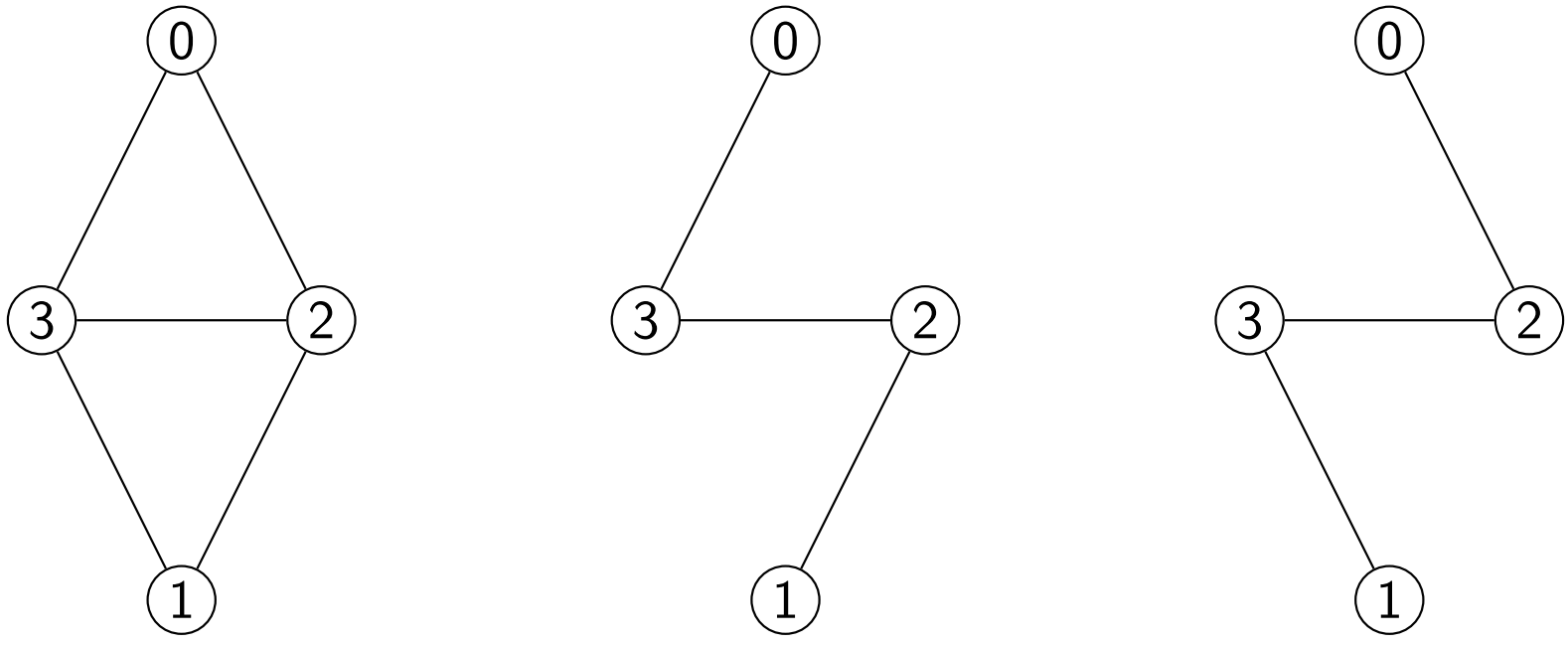


(e) A root of K_6.



(f) A root of K_7.

Figure: A few of the 15 non-isomorphic graph roots of K_5.



(a) G (b) T_1 (c) T_2

Figure: The graph G has two possible P_4 square roots that are trees, T_1 and T_2. The two roots are isomorphic, differing only by the interchange of vertices 2 and 3.

NP-Completeness

Motwani and Sudan (1994) proved that the graph square problem is **NP**-complete by showing that the problem is **NP**-hard through a polynomial reduction from Not-All-Equal 3-SAT.

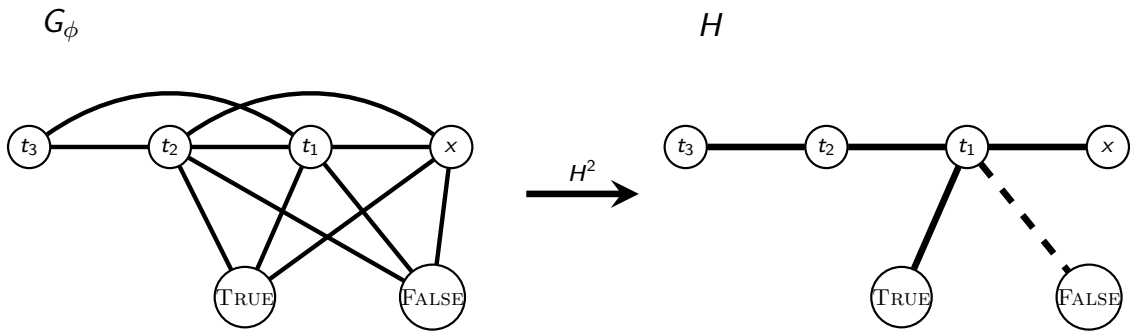
Problem (NAESAT)

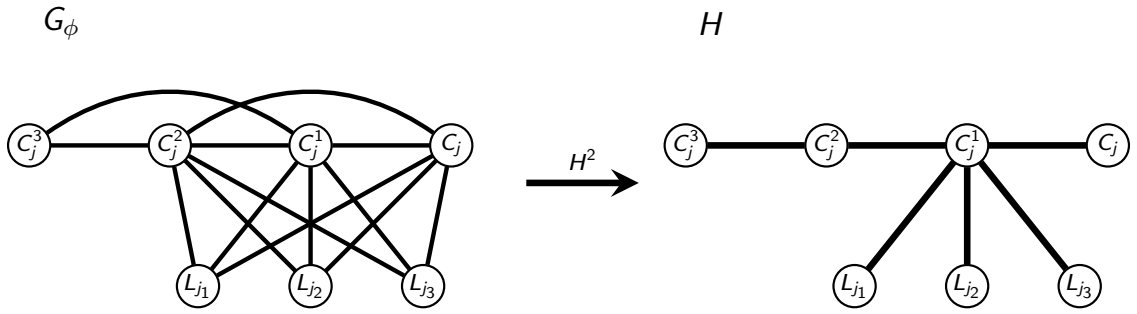
Instance: *A 3-CNF formula*

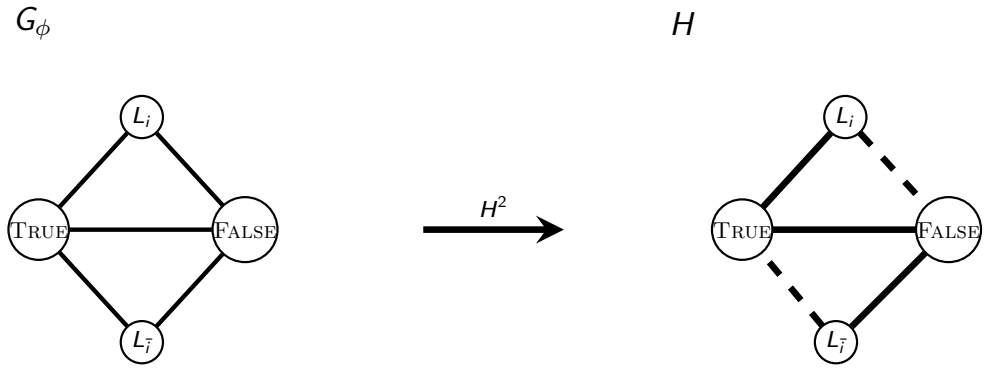
$$\phi = C_1 \wedge C_2 \wedge \cdots \wedge C_m$$

where each clause C_i is over variables $x_1, \ldots, x_n$.

Question: *Does there exist an assignment of* TRUE *and* FALSE *to the variables such that every clause contains at least one* TRUE *literal and at least one* FALSE *literal?*


G_ϕ
H
t_3
t_2
t_1
x
True
False
H^2


G_ϕ
H
C_j^3
C_j^2
C_j^1
C_j
L_{j_1}
L_{j_2}
L_{j_3}
H^2


G_ϕ
H
L_i
True
False
$L_{\bar{i}}$
H^2

Restricting the Square vs the Root

Square G	**Root H**	**Main result**
Unrestricted	Trees	Ross and Harary (1960): structural characterization and reconstruction of tree roots, unique up to isomorphism [RH60].
Unrestricted	Trees	Lin and Skiena (1995): optimal linear-time recognition and reconstruction of tree roots [LS95].
2-connected	Unrestricted	Fleischner (1974): the square of a 2-connected graph has a Hamiltonian cycle [Fle74].
Unrestricted	Interval squares	Lau and Corneil (2004): polynomial-time construction of roots from interval squares [LC04].
Unrestricted	Block graphs	Le and Tuy (2007): linear-time recognition of roots of block graphs [LT07].
Unrestricted	Bipartite	Lau (2006): linear-time construction of bipartite roots [Lau06].
Directed graphs	Directed graphs	Geller (1970): necessary and sufficient conditions for directed square-root existence [Gel70].
Unrestricted	Graphs of prescribed girth	Farzad, Lau, Le, and Tuy (2009) and Farzad and Karimi (2012): complexity dichotomy for square-root recognition [Far+09; FK12].
Maximum degree ≤ 6	Unrestricted	Cochefert, Couturier, Golovach, Kratsch, and Paulusma (2013): minimum-edge roots in polynomial time and maximal-edge roots in exponential time [Coc+13].
Unrestricted	Oriented graphs, maximum degree ≤ 3, girth ≥ 4	Duffy (2025): classification of squares of oriented graphs under these root restrictions [Duf25].

Maximal Cliques and Unicliqual Vertices in Tree Roots

Input: A graph G where $G = T^2$
Output: A tree T
Step 1
if $G = K_p$ is complete **then**
    Return tree T with depth 1, root any vertex
else
    Continue
end if
Step 2
Identify all maximal cliques $C_1, \ldots, C_n$ of G (e.g. Bron–Kerbosch [BK73])
Step 3
Let $b_1, \ldots, b_n$ be multicliqual points for $C_1, \ldots, C_n$
Build graph T on $\{b_1, \ldots, b_n\}$
for each pair (C_i, C_j) with $|C_i \cap C_j| = 2$ **do**
    Add edge (b_i, b_j) in T
end for
T is a tree
Step 4
for each clique C_i **do**
    Let n_i be the number of unicliqual points in C_i
    Attach n_i leaves to b_i in T
end for
Resulting T has $|V(G)|$ nodes and $T^2 = G$

Integer Program

Two distinct vertices are adjacent in $G = H^2$ if and only if they are connected by a path of length one or two in H.

$$A_G = (A_H)^2 \vee A_H. \tag{2}$$

$$(A_H)^2_{ij} = \big((A_H)_{i1} \wedge (A_H)_{1j}\big) \vee \big((A_H)_{i2} \wedge (A_H)_{2j}\big) \vee \cdots \vee \big((A_H)_{in} \wedge (A_H)_{nj}\big)$$

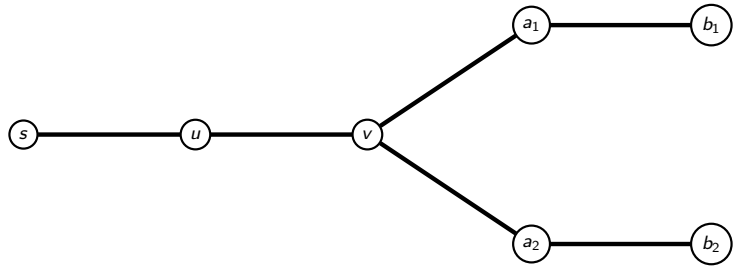
$$g_{ij} \text{ if and only if } \Big(\bigwedge_k (x_{ik} \wedge x_{kj}) \vee x_{ij}\Big). \tag{3}$$

Lemma

Let G and H be graphs with adjacency matrices A_G and A_H,respectively. The graph H is a root of G if and only if

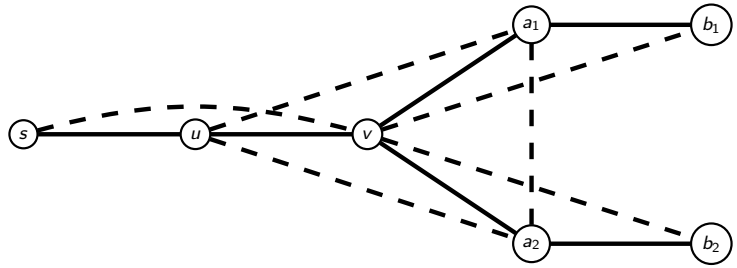
$$A_G = \mathbb{1}_{\geq 1}\big((A_H)^2 - D_H + A_H\big).$$

Vertices ordered $s, u, v, a_1, a_2, b_1, b_2$.



$$A_H = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 \end{pmatrix} \qquad (A_H)^2 = \begin{pmatrix} 1 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 2 & 2 & 0 & 0 \\ 1 & 0 & 3 & 0 & 0 & 2 & 2 \\ 0 & 2 & 0 & 2 & 1 & 0 & 0 \\ 0 & 2 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 1 \end{pmatrix}$$

$$D_H = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$



$$A_H^2 - D_H + A_H = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 2 & 2 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 & 1 & 1 & 0 \\ 0 & 2 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix} \mathbb{1}_{\geq 1}\left(A_H^2 - D_H + A_H\right) = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

$$A_G = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 & 0 \end{pmatrix}$$

i) $x_{ij} \in \{0,1\}$ denotes the (i,j) entry of the adjacency matrix A_H.

ii) $y_{ikj} \in \{0,1\}$ represents the product $x_{ik}x_{kj}$ in $(A_H)^2_{ij}$.

iii) $z_{ij} \in \mathbb{Z}_{\geq 0}$ represents the argument of the indicator function.

iv) $M \in \mathbb{Z}_{>0}$ be a sufficiently large constant $M \leq n(n-1) = O(n^2)$.

$$D_H = \begin{bmatrix} \deg_H(1) & 0 & \cdots & 0 & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ \vdots & \cdots & \deg_H(i) & \cdots & 0 \\ 0 & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & \deg_H(n) \end{bmatrix}, \quad A_H = \begin{bmatrix} x_{11} & \cdots & x_{1j} & \cdots & x_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \cdots & x_{ij} & \cdots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{n1} & \cdots & x_{nj} & \cdots & x_{nn} \end{bmatrix}$$

$$(A_H)^2 = \begin{bmatrix} \sum_k x_{1k}x_{k1} & \cdots & \sum_k x_{1k}x_{kj} & \cdots & \sum_k x_{1k}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{ik}x_{k1} & \cdots & \sum_k x_{ik}x_{kj} & \cdots & \sum_k x_{ik}x_{kn} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \sum_k x_{nk}x_{k1} & \cdots & \sum_k x_{nk}x_{kj} & \cdots & \sum_k x_{nk}x_{kn} \end{bmatrix}.$$

$$\begin{bmatrix} g_{11} \\ \vdots \\ g_{1j} \\ \vdots \\ g_{1n} \\ \vdots \\ g_{ij} \\ \vdots \\ g_{nn} \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}(z_{11}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1j}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{1n}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{ij}) \\ \vdots \\ \mathbb{1}_{\geq 1}(z_{nn}) \end{bmatrix} = \begin{bmatrix} \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{k1} - (D_H)_{11} + x_{11}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kj} - (D_H)_{1j} + x_{1j}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{1k}x_{kn} - (D_H)_{1n} + x_{1n}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{ik}x_{kj} - (D_H)_{ij} + x_{ij}\big) \\ \vdots \\ \mathbb{1}_{\geq 1}\big(\sum_{k=1}^{n} x_{nk}x_{kn} - (D_H)_{nn} + x_{nn}\big) \end{bmatrix}. \tag{4}$$

Objective Function:

$$\max \quad 0 \tag{5}$$

Variables:

$$x_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{6}$$

$$y_{ikj} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{7}$$

$$z_{ij} \in \mathbb{Z}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{8}$$

Constraints:

(1) Indicator argument:

$$z_{ii} = \sum_{k=1}^{n} y_{iki} - \sum_{k=1}^{n} x_{ik}, \forall i \in \{1, \ldots, n\}, \tag{9}$$

$$z_{ij} = \sum_{k=1}^{n} y_{ikj} + x_{ij}, \qquad \forall i \neq j \in \{1, \ldots, n\}, \tag{10}$$

(2) Indicator function:

$$z_{ij} \leq M g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{11}$$

$$z_{ij} \geq g_{ij}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{12}$$

(3) Product:

$$y_{ikj} \leq x_{ik}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{13}$$

$$y_{ikj} \leq x_{kj}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{14}$$

$$y_{ikj} \geq x_{ik} + x_{kj} - 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{15}$$

(4) Upper Bound:

$$x_{ij} \leq 1, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{16}$$

$$y_{ikj} \leq 1, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{17}$$

(5) Symmetry:

$$x_{ij} = x_{ji}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{18}$$

(6) Diagonal:

$$x_{ii} = 0, \qquad \forall i \in \{1, \ldots, n\}. \tag{19}$$

The integer program on the arbitrary graph G using Equations (6)–(19) together with the constant objective function Equation (5) is denoted as $\mathcal{IP}(G)$.
The total number of simple graphs on $n = 2, 3, 4, \ldots$ unlabeled nodes is the sequence $2, 4, 11, 34, 156, 1044, 12346, \ldots$ given by the OEIS sequence `A000088`. The number of graphs which are squares is the sequence $2, 3, 6, 11, 28, 77, 307, \ldots$ given by `A382181`.

n	H exists	H does not exist
3	3	1
4	6	5
5	11	23
6	28	128
7	77	967
8	307	12039

Table: Outcomes of the integer program for graphs of different orders of n.

Methodology

Implemented in **SageMath**, with **Gurobi** as the optimization backend.

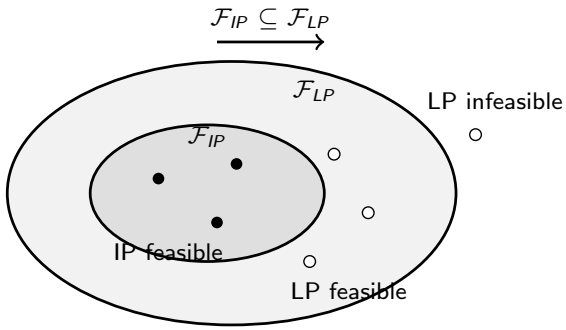
Uses the **barrier method**, an interior-point method based on predictor-corrector Newton iterations.

Results correspond to the labelled representatives produced by the specified generation procedure.

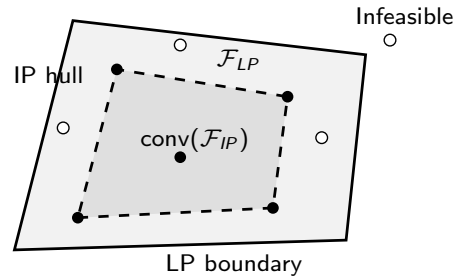
Code and test harness: `https://github.com/Hellwoodoo/Graph-Squares`.

```
def linear_program(G):
    m = MixedIntegerLinearProgram(solver='Gurobi', maximization=False)
    m.solver_parameter("Method", 2)
    x = m.new_variable(real=True, nonnegative=True)
    y = m.new_variable(real=True, nonnegative=True)
    z = m.new_variable(real=True, nonnegative=True)
    A_G = G.adjacency_matrix()
    n = A_G.nrows()
    M = n**2
    for i in range(n):
        for j in range(n):
            if i == j:
                m.add_constraint(z[i,j] - (sum(y[i,k,j] for k in range(n)) - sum(x[i,k] for k in range(n)) + x[
    i,j]) == 0)
            else:
                m.add_constraint(z[i,j] - (sum(y[i,k,j] for k in range(n)) + x[i,j]) == 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(z[i,j] - M * A_G[i][j] <= 0)
            m.add_constraint(-z[i,j] + A_G[i][j] <= 0)
    for i in range(n):
        for j in range(n):
            for k in range(n):
                m.add_constraint(y[i,k,j] - x[i,k] <= 0)
                m.add_constraint(y[i,k,j] - x[k,j] <= 0)
                m.add_constraint(- y[i,k,j] + x[i,k] + x[k,j] - 1 <= 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(x[i,j] == x[j,i])
    for i in range(n):
        m.add_constraint(x[i,i] == 0)
    for i in range(n):
        for j in range(n):
            m.add_constraint(x[i,j] - 1 <= 0)
            for k in range(n):
                m.add_constraint(y[i,k,j] - 1 <= 0)
    return m, x
```

Linear Relaxation



Set-theoretic



Geometric

$$x_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}, \tag{20}$$

$$y_{ikj} \in \mathbb{R}_{\geq 0}, \qquad \forall i, k, j \in \{1, \ldots, n\}, \tag{21}$$

$$z_{ij} \in \mathbb{R}_{\geq 0}, \qquad \forall i, j \in \{1, \ldots, n\}. \tag{22}$$

Such a linear program may have *integral solutions* and *fractional solutions* when the root exists and *fractional solutions* and *infeasible solutions* when the root does not exist.

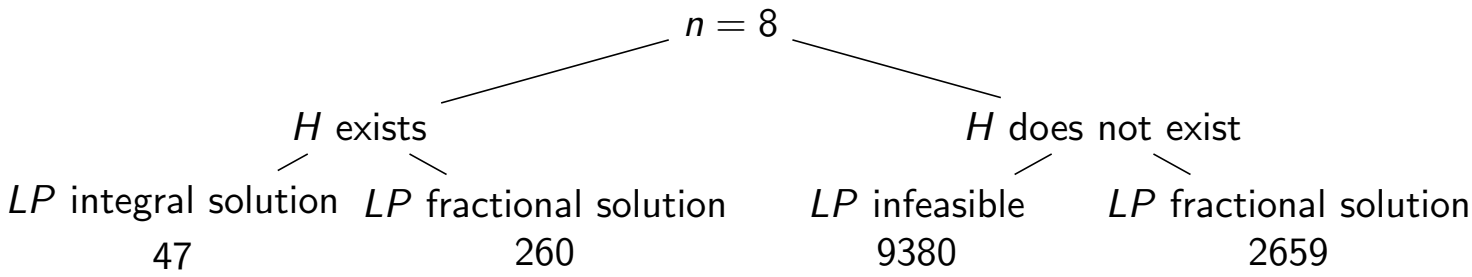


Figure: LP outcomes for $n = 8$ vertices using $\mathcal{LP}(G)$.

The LP integral solution sequence $2, 3, 6, 10, 17, 29, 51, \ldots$ matches A285665. However, it is unclear why this should be the case.

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	5	1	5	0
5	7	4	20	3
6	13	15	107	21
7	22	55	767	200
8	44	263	9380	2659

Table: Outcomes of the LP relaxation for graphs of different orders n, classified according to whether a square root H exists.

The linear relaxation will never return an infeasible solution if H exists.

Proposition

If G has a square root H, then $\mathcal{LP}(G)$ is feasible.

Proof.

Since $\mathcal{LP}(G)$ is obtained by relaxing the integrality constraints of $\mathcal{IP}(G)$, every feasible solution of $\mathcal{IP}(G)$ is also feasible for $\mathcal{LP}(G)$. Hence $\mathcal{LP}(G)$ is feasible. □

Min and Max Objective Functions

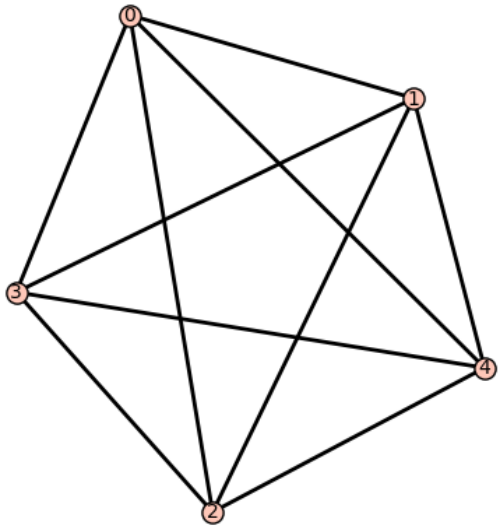
$$\min \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}$$

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	2	1	1	0
4	3	3	5	0
5	3	8	20	3
6	4	24	107	21
7	5	72	767	200
8	7	300	9380	2659

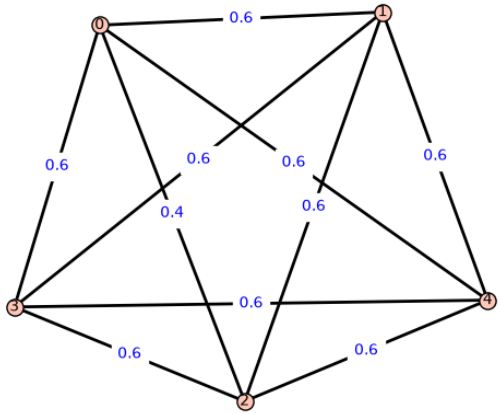
$$\max \sum_{i=1}^{n} \sum_{j=1}^{n} x_{ij}$$

n	H exists		H does not exist	
	LP integral solution	LP fractional solution	LP infeasible	LP fractional solution
3	3	0	1	0
4	6	0	5	0
5	9	2	20	3
6	18	10	107	21
7	38	39	767	200
8	94	213	9380	2659

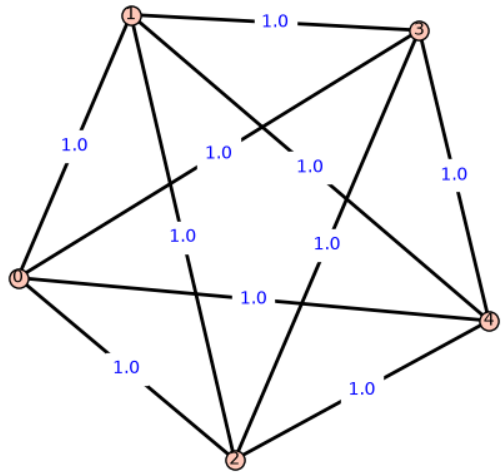
The single complete graph K_5 is selected as the unique optimal solution.



(a) K_5.



(b) Linear program result for K_5.



(c) Max linear program result for K_5.

Figure: The graph K_5 and two of its square roots.

Fundamental Properties

Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have $x_{ab} = y_{avb} = 0$.

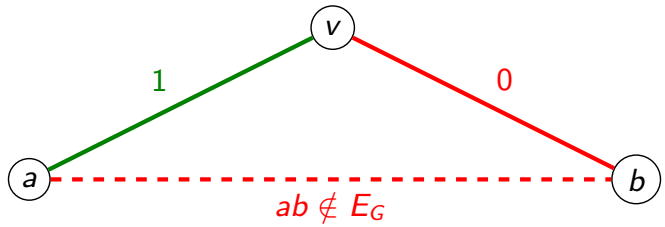
Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\mathcal{LP}(G)$ and every $v \in V_G$ with $v \neq a, b$ we have

$$x_{av} + x_{vb} \leq 1.$$

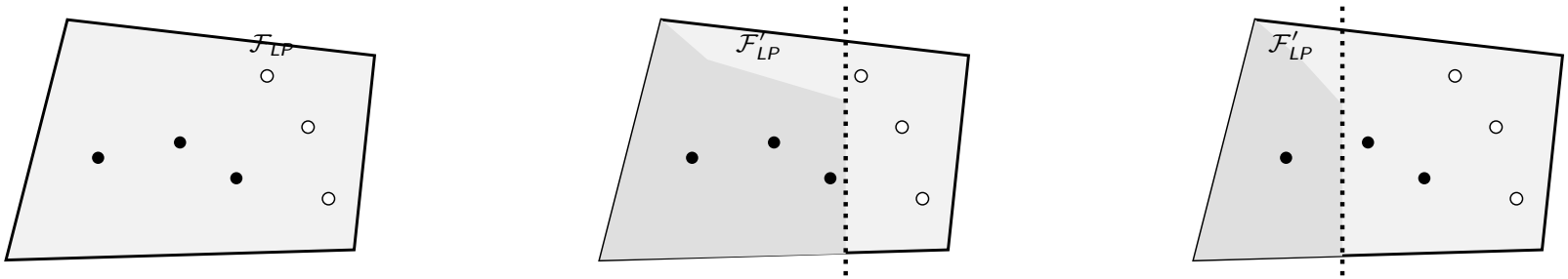
Lemma

Let G and H be graphs such that $G = H^2$. Let $a, b, v \in V_G$ such that $ab \notin E_G$. If a feasible solution x of $\mathcal{LP}(G)$ satisfies $x_{av} = 1$, then $x_{vb} = 0$.



Strengthening the LP

Goal: build a new LP $\max \mathcal{UVLP}(G)$ by adding constraints that let all tree roots be identified.



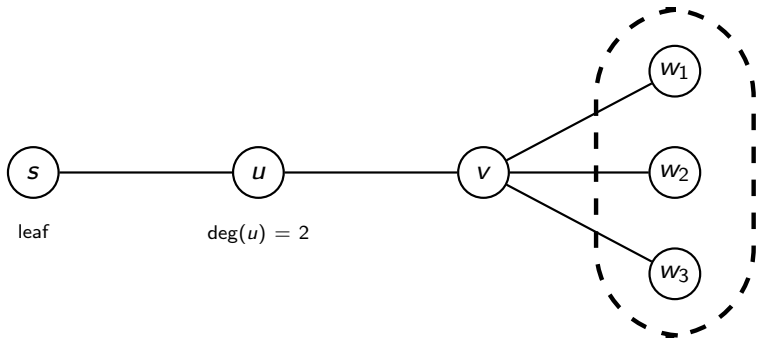
Base LP **Useful strengthening** **Incorrect strengthening**

A suitable constraint eliminates fractional solutions while retaining all valid integer solutions, whereas an overly restrictive constraint also excludes valid integer solutions.

Identifying Square Characteristics of Tree Roots

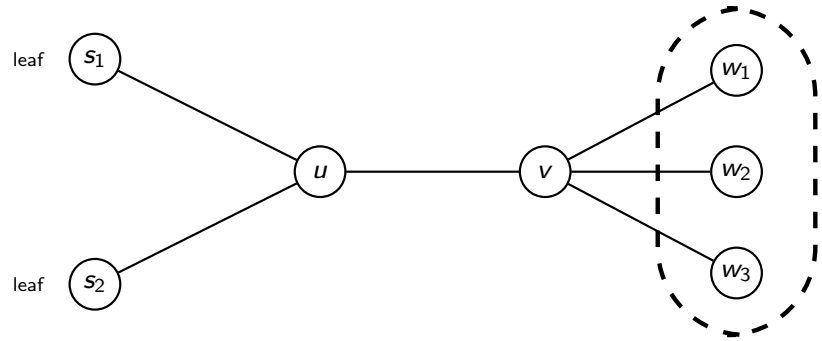
Definition (Sparse Pendant)

Let H be a graph. A **sparse pendant** is a path $s - u - v$ in H such that s is a leaf and u has degree 2.



Definition (Pendant Fork)

Let H be a graph. A **pendant fork** is the induced subgraph of H on four vertices s_1, s_2, u, v, where s_1 and s_2 are leaves adjacent to u, and every neighbour of u other than v is a leaf.



Sparse Pendants or Pendant Forks in Trees

Lemma

Every non-trivial tree with at least 3 vertices contains either a sparse pendant or a pendant fork subgraph.

Proof.

Root T at any vertex. Let s be a leaf of maximum depth, u its parent. Every child of u is a leaf, else a deeper vertex than s would exist. **Case 1: u is not the root.** Let v be the parent of u. Every neighbour of u other than v is a leaf child.

If u has exactly one leaf neighbour s, then $\deg_T(u) = 2$, and $\{s, u, v\}$ induces a sparse pendant.

If u has two leaf neighbours s_1, s_2, then $\{s_1, s_2, u, v\}$ induces a pendant fork.

Case 2: u is the root. All neighbours of u are leaf children, and u has at least two of them.

If u has exactly two neighbours s, v, then $\{s, u, v\}$ induces a sparse pendant.

Otherwise, pick distinct children s_1, s_2, v of u; then $\{s_1, s_2, u, v\}$ induces a pendant fork.

Leaf Fixing Linear Program

Using Lemma (9) we can identify s to be a leaf, u its corresponding stem and vertex v to give the fix

$$x_{su} = x_{us} = x_{uv} = x_{vu} = 1. \tag{23}$$

The construction of the linear program on the arbitrary graph G using $\max \mathcal{LP}(G)$ with the addition of Equation (**??**) is denoted as $\max \mathcal{LP}(G)$.

Lemma

Let G and T be graphs such that $G = T^2$ and T is a tree. $\max \mathcal{UVLP}(G)$ *always fixes a sparse pendant or pendant fork subgraph of T.*

Proposition (Validity of Unicliqual Vertex Fix)

If a root H of G exists, then $\max \mathcal{UVLP}(G)$ *is feasible.*

$\max \mathcal{LP}$

n	H exists		H does not exist	
	integral	fractional	infeasible	fractional
3	3	0	1	0
4	6	0	5	0
5	10	1	21	2
6	22	6	117	11
7	51	26	852	115
8	148	159	10271	1768

$\max \mathcal{UVLP}$

n	H exists		H does not exist	
	integral	fractional	infeasible	fractional
3	3	0	1	0
4	6	0	5	0
5	9	2	20	3
6	18	10	107	21
7	38	39	767	200
8	94	213	9380	2659

Theorem

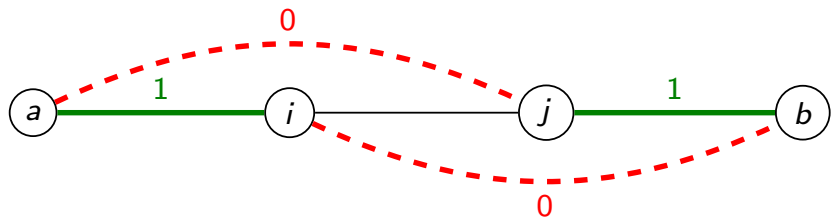
Let G and T be graphs such that $G = T^2$ and T is a tree. Then $\mathcal{UVLP}(G)$ returns an integral solution.

Propagation of the Fixing

Lemma (0-Path Propagation)

Let G and H be graphs such that $G = H^2$. Let $a, i, j, b \in V_G$ such that a, i, j, b form a shortest induced path in H between a and b. For every feasible solution x of $\mathcal{LP}(G)$,

$$x_{ai} = 1 \implies x_{ib} = 0, \quad x_{ia} = 1 \implies x_{bi} = 0, \quad x_{jb} = 1 \implies x_{aj} = 0, \quad x_{bj} = 1 \implies x_{ja} = 0.$$



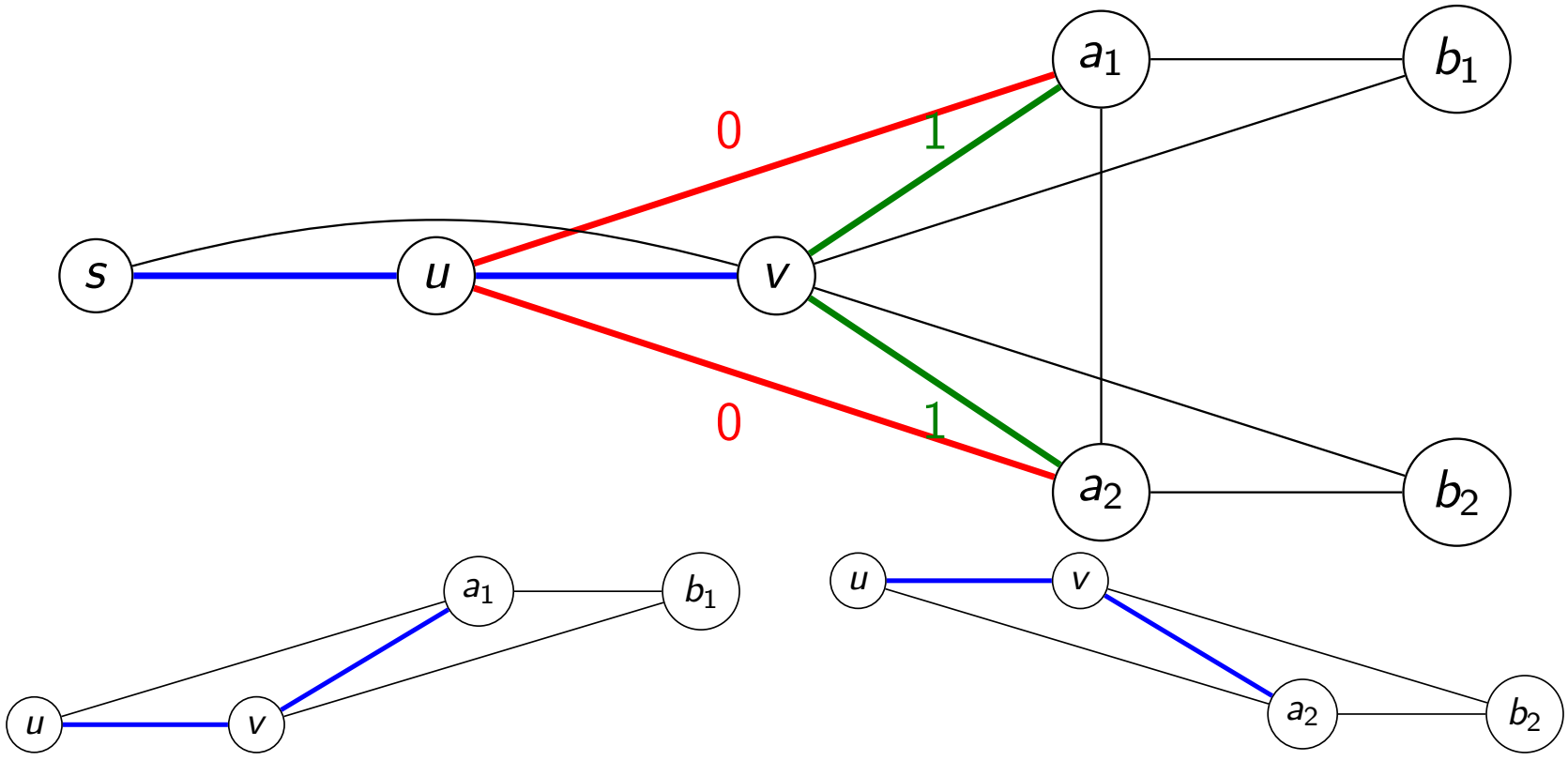
Lemma (1-Path Propagation)

Let G be a graph and let x be a feasible solution of $\mathcal{LP}(G)$. Suppose that $u - v - a_1$ form a shortest induced path in H between u and a_1 and that $x_{uv} = 1$. Let $A = \{a_1, \ldots, a_r\}$ and $B = \{b_1, \ldots, b_s\}$ be such that every vertex $k \in V_G \setminus \{u, v\}$ satisfies $k \in A \cup B$, and suppose that

$$x_{um} = 0 \text{ for every } m \in A, \quad g_{un} = 0 \text{ for every } n \in B,$$

and $g_{sa_1} = 0$ for the vertex s preceding u on the shortest path to a_1. Then

$$x_{va_1} = 1.$$


a_1
b_1
0
1
s
u
v
0
1
a_2
b_2
a_1
b_1
u
v
u
v
a_2
b_2

Symbolic Sets

Conjecture

Let G and H be graphs such that $G = H^2$. Let $a, b \in V_G$ such that $ab \notin E_G$. For every feasible solution x of $\max \mathcal{LP}(G)$ *and every $v \in V_G$ with $v \neq a, b$ we have*

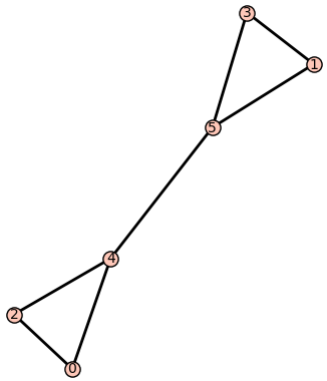
$$x_{av} + x_{vb} = 1 \text{ or } x_{av} + x_{vb} = 0.$$

Definition (Consistent Set)

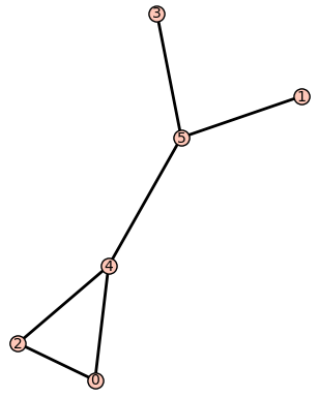
Take a maximal set $E' \subseteq E_G$ of at least three edges in which every edge is linked to another edge of E' by the $= 1$ relationship. We call E' **consistent** if its edges can be split into two nonempty groups A and B so that no two edges within A form a P_3 in G, and the same holds within B.

Definition (Inconsistent Set)

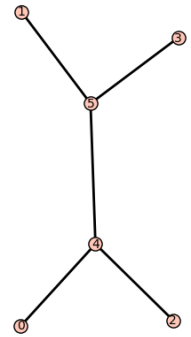
Let E' be as above. We call E' **inconsistent** if no such split exists. That is, if we divide its edges into two nonempty groups A and B, at least one of the groups contains two edges forming a P_3 in G.



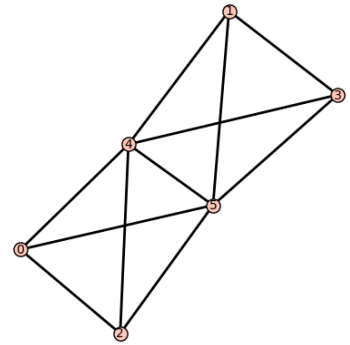
(a) The root graph G_1.



(b) The root graph G_2.



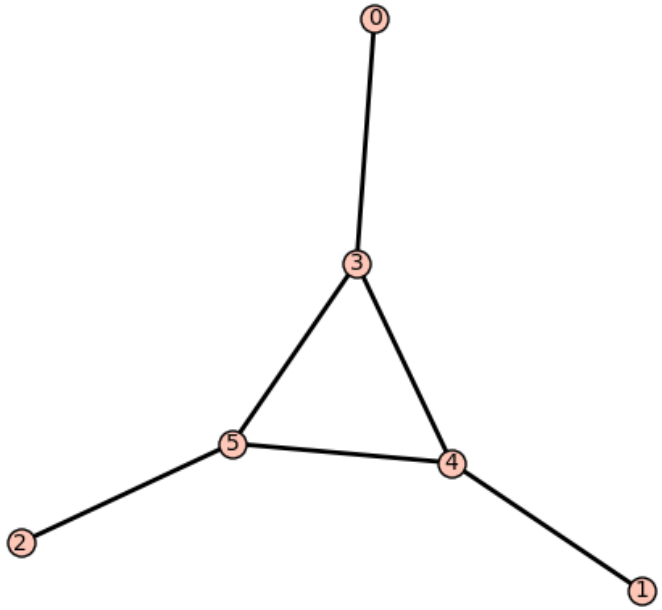
(c) The root graph G_3.



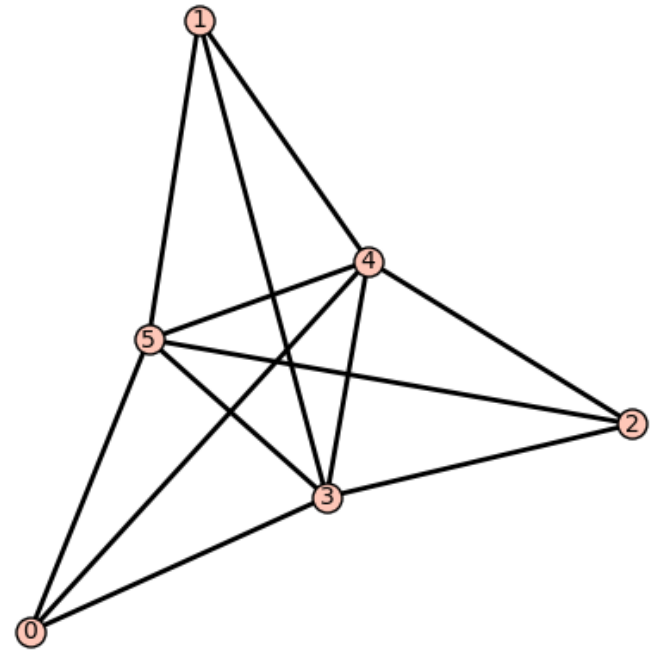
(d) The graph square $H = G_i^2$.

Figure: Three non-isomorphic square roots of the same six-vertex graph square.

$\{(1,4),(4,0),(4,2),(4,3)\}$ and $\{(1,5),(3,5),(5,0),(5,2)\}$ are consistent sets



(a) The root graph G.



(b) The graph square H such that $H^2 = G$.

Figure: A six-vertex graph and its square. G only has one root, although there are different relabellings.

$\{(0,3),(3,1),(3,2)\}$, $\{(0,5),(1,5),(5,2)\}$ and $\{(4,2),(4,1),(4,0)\}$ are inconsistent sets

max $\mathcal{LP}$

n	H exists		H does not exist	
	integral	fractional	infeasible	fractional
3	3	0	1	0
4	6	0	5	0
5	10	1	21	2
6	22	6	117	11
7	51	26	852	115
8	148	159	10271	1768

Edge enforcement (base: max $\mathcal{LP}$**)**

n	H exists		H does not exist	
	integral	fractional	infeasible	fractional
3	3	0	1	0
4	6	0	5	0
5	10	1	20	3
6	26	2	107	21
7	67	10	767	200
8	240	67	9380	2659

Outcomes of the greedy iterative edge enforcement linear program using the max linear program as the base for graphs of different orders of n as compared to the original max linear program.

Rounding

1: **Input:** A graph G
2: **Output:** A LP model (m, x)
3: Initialize $F = \emptyset$ and construct $\mathcal{UVLP}(G)$.
4: Solve $\mathcal{UVLP}(G)$.
5: **while** the current LP solution is fractional **do**
6: For each fractional edge ab, compute its rounding score $s_{ab} = |x_{ab} - 0.5|$.
7: Select a fractional edge ab with maximum score.
8: Set

$$\widehat{x}_{ab} = \widehat{x}_{ba} = \begin{cases} 0, & x_{ab} < 0.5, \\ 1, & x_{ab} \geq 0.5. \end{cases}$$

9: Add ab with its assigned value to F.
10: Reconstruct $\mathcal{UVLP}(G)$ with all fixings in F.
11: Solve $\mathcal{UVLP}(G) + F$.
12: **if** $\mathcal{UVLP}(G) + F$ is infeasible **then**
13: **return** the previous feasible solution (m, x).
14: **end if**
15: **if** $\mathcal{UVLP}(G) + F$ is integral **then**
16: **return** the integral solution (m, x).
17: **end if**
18: **end while**
19: **return** (m, x).

n	H exists		H does not exist		Integral $\leftarrow$ Fractional
	LP integral	LP fractional	LP infeasible	LP fractional	
3	3	0	1	0	0
4	6	0	5	0	0
5	11	0	20	3	2
6	27	1	107	21	9
7	73	4	767	200	35
8	245	62	9380	2659	151

Table: Outcomes of the greedy rounding linear program using the max linear program as the base for graphs of different orders of n.

n	H exists		H does not exist		Integral $\leftarrow$ Fractional
	LP integral	LP fractional	LP infeasible	LP fractional	
3	3	0	1	0	0
4	6	0	5	0	0
5	11	0	21	2	1
6	28	0	117	11	6
7	76	1	852	115	25
8	268	39	10271	1768	120

Table: Outcomes of the greedy rounding linear program using the unicliqual vertex linear program as the base for graphs of different orders of n.